Price Determinants in the NYC Airbnb Market: A Machine Learning Study

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1. Introduction and Background

The rise of digital platforms has transformed the hospitality industry. Airbnb has emerged as one of the leading facilitators of short-term rentals across major cities and tourist destinations around the world. In the highly competitive and dynamic market of New York City, Airbnb listing prices vary significantly due to various factors, including geographic location, property type, and listing availability throughout the year. Airbnb hosts must strategically juggle pricing their properties to maximize revenue while also attracting potential guests, and travelers must rely on transparent pricing to make informed booking decisions. As pricing is one of the most critical determinants of consumer behavior and market performance, understanding how key listing features influence nightly rental rates provides valuable economic and strategic insights to renters, tenants, and local policymakers.

This project uses real Airbnb data from New York City to show how listing prices and rental types are related using linear regression. We aim to analyze core features—such as borough, neighborhood, room type, and annual availability—to determine which factors most significantly influence rental price variations across the city. This analysis will elucidate practical ramifications for property owners, travelers, and investors by enhancing comprehension of Airbnb pricing trends and suggesting potential pricing strategies. Conversely, the findings will also allow customers to identify cost-effective accommodation options and provide policymakers with insight into how short-term rentals impact local housing and tourism markets.

2. Research Motivation

Understanding the factors that drive Airbnb pricing is important because Airbnb plays a significant role in tourism and housing markets worldwide. Airbnb has changed how people travel, how property owners make income, and the availability of housing for residents.

It can be difficult for hosts to determine how to set competitive pricing while maximizing potential earnings. Many hosts set pricing based on intuition rather than evidence, which causes costs to be inconsistent across the Airbnb website. Our goal is to support hosts by providing information to maximize revenue while receiving an optimal number of bookings.

As Airbnb rental is continuously growing as a preferred accommodation option over hotels, it can be difficult for travelers to know what factors to look for when they want affordability. There are large pricing differences between rentals, a lack of predictability, and a lack of transparency from hosts. This analysis aims to provide guidance to Airbnb customers on the listing characteristics that most influence. This analysis will also explore pricing strategies and methods for finding more affordable rental options. This study focuses on the strategies employed to find more affordable rental options and strategies for locating more affordable rentals. Additionally, this analysis aims to convince renters to pursue more evidence-based pricing on rentals.

Platforms for short-term rentals—such as Airbnb—also influence housing availability and rent pressure in neighborhoods. Airbnb is responsible for an approximate 9.2 percent increase in citywide rental rates in New York City alone (2). This analysis can provide policymakers with better insight into home pricing in tourist-heavy areas and show the impact short-term rentals have on the local housing market.

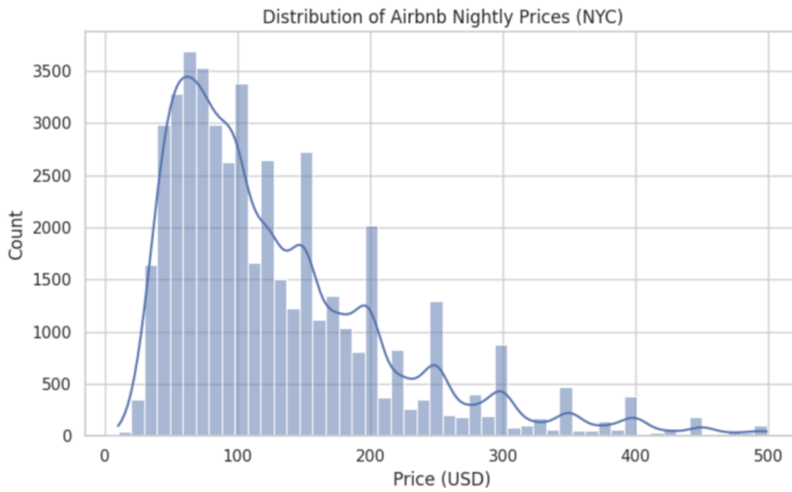
3. Data and Variables

The dataset used for this project is the *New York City Airbnb Open Data*, sourced from Kaggle and originally compiled by *Inside Airbnb* (1). It contains approximately 49,000 listings, with each listing containing 16 variables and with each row representing active Airbnb properties available for short-term rentals in New York City. The dataset includes 16 variables: five integer datatypes, five string datatypes, two identifiers, and 4 miscellaneous datatypes. For the purposes of our model, we were only interested in the variables “price,” “neighborhood_group,” “neighborhood,” “room_type,” and “availability.”

4. Descriptive Statistics

To summarize our dataset, our team utilized seven methods: a histogram, a boxplot of Airbnb prices by borough, a boxplot of Airbnb prices by room type, a summary table describing the numeric columns in the data, a summary table describing the average price by borough and room type, a scatterplot, and a heatmap. We describe our findings from each method below.

Histogram

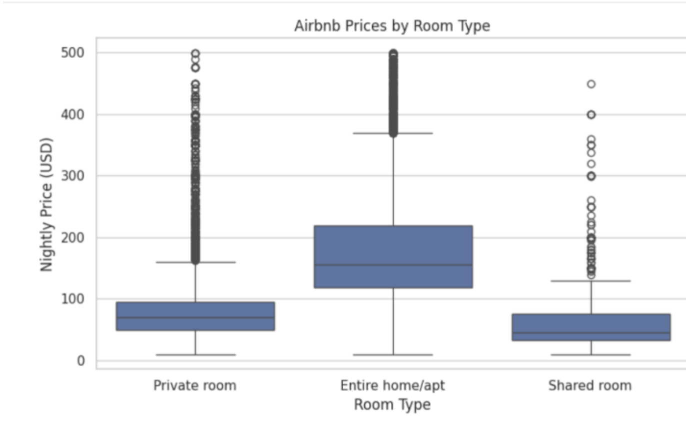


We can see that the price distribution is right-skewed, indicating that most listings are mid-to-low priced and only a few are very expensive.

Boxplot – Prices by Borough

We can see that the median price in Manhattan is the highest, with the outer boroughs showing lower prices and more variation.

Boxplot – Prices by Room Type



We can see that entire house/apartment listings are much more expensive than private rooms or shared rooms.

Summary Table – Describe Numeric Columns

	price	minimum_nights	number_of_reviews	reviews_per_month	calculated_host_listings_count	availability_365
count	47649.000000	47649.000000	47649.000000	38097.000000	47649.000000	47649.000000
mean	130.083926	6.978006	23.591492	1.377636	7.096812	111.016768
std	85.074554	20.339170	44.871748	1.684888	32.825837	130.879436
min	10.000000	1.000000	0.000000	0.010000	1.000000	0.000000
25%	68.000000	1.000000	1.000000	0.190000	1.000000	0.000000
50%	100.000000	2.000000	5.000000	0.720000	1.000000	42.000000
75%	170.000000	5.000000	24.000000	2.030000	2.000000	221.000000
max	499.000000	1250.000000	629.000000	58.500000	327.000000	365.000000

We see a basic statistical summary of numerical variables (price, minimum nights, number of reviews, etc.) in the dataset.

Summary Table – Average Price by Borough and Room Type

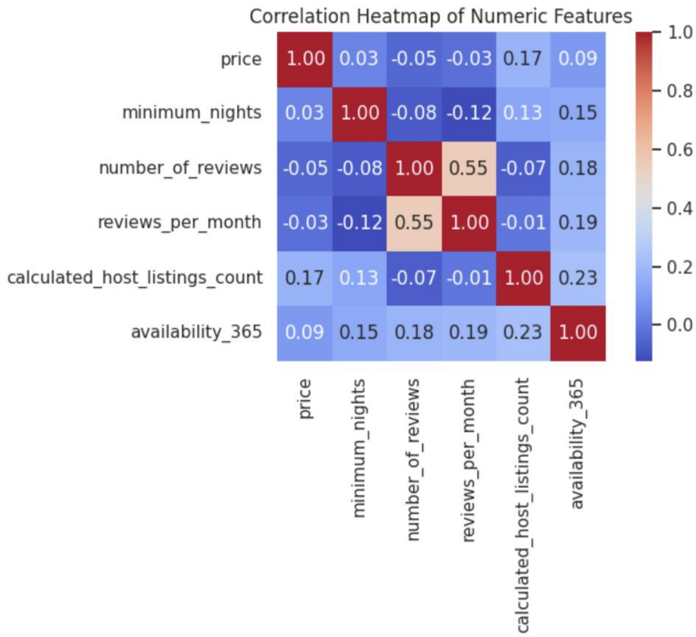
	neighbourhood_group	room_type	price
6	Manhattan	Entire home/apt	201.30
3	Brooklyn	Entire home/apt	158.30
9	Queens	Entire home/apt	137.88
12	Staten Island	Entire home/apt	124.84
0	Bronx	Entire home/apt	121.48
7	Manhattan	Private room	102.44
8	Manhattan	Shared room	82.33
4	Brooklyn	Private room	71.16
10	Queens	Private room	66.89
13	Staten Island	Private room	62.29

We see that across all boroughs, the "entire home/apartment" room type consistently showed the highest prices. Manhattan stands out as the borough with the highest average prices overall.

Scatterplot

We can see that there is no clear linear relationship between price and availability, indicating that the price is not determined by availability alone.

Heatmap



In this heatmap, we are visualizing correlations between each numeric feature and price. We can see that "calculate_host_listings_count" has the strongest positive correlation to price (0.17) among numeric variables.

5. Data Mining Model Methodology

Performing linear regression allows us to assess how each of the variables we identified for analysis - "neighborhood_group," "neighborhood," "room_type," and "availability" - contributes to Airbnb price variations. The model provides us with coefficients that show the direction and strength of each variable's influence. This method is particularly effective for our analysis because we want to quantify each variable's impact. By understanding the driving forces behind Airbnb pricing and utilizing the model to predict future prices, the relationship between price and features becomes clear and measurable, allowing for important conclusions about the magnitude of factors influencing Airbnb pricing.

Linear regression is also appropriate in that its assumptions align well with our data, where relationships between variables can be reasonably modeled as additives. Additionally, linear regression shows how each variable affects the price without relying on complex black-box models. This interpretability further strengthens the practical value of our analysis in that our goal is not just to predict prices but also to understand the key factors that determine those prices.

In addition to linear regression analysis, a random forest regression model was applied to explore potential nonlinear relationships among variables. Random forest is an ensemble technique that combines multiple decision trees and is capable of capturing complex interactions and nonlinear effects that may exist between location characteristics, accommodation type, and availability variables. By comparing the performance of the linear regression model and the Random Forest model, we evaluated whether increased model flexibility leads to improved Airbnb price prediction accuracy.

Furthermore, an XGBoost regression model was applied to further enhance predictive performance. XGBoost is a gradient boosting–based approach that sequentially learns decision trees while correcting errors from previous iterations. This method is particularly effective for structured data and is well-suited for capturing localized and nonlinear price patterns, especially at finer geographic levels such as individual neighborhoods. In this analysis, we examined whether XGBoost can more precisely model the complexity of Airbnb price formation.

By using linear regression, Random Forest, and XGBoost models together, we are able to compare different levels of model complexity and interpretability. This multi-model approach not only identifies the best-performing predictive model, but also provides a comprehensive understanding of whether Airbnb price fluctuations are driven primarily by linear relationships or whether nonlinear factors play a significant role.

6. Data Mining Model Evaluation

6.1. Linear Regression

To conduct our analysis, we first utilized a linear regression model with an emphasis on predicting rental prices based on continuous factors (borough, neighborhood, room type, and availability). The dependent variable in our analysis was "price," and the independent variables were "neighborhood group," "neighborhood," "room type," and "availability."

Linear Regression Model Performance

```
MAE: 43.1088667235697
MSE: 3952.4658696092733
RMSE: 62.86863979448954
R-Squared: 0.4518845955854657
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The linear regression model achieved a Mean Absolute Error (MAE) of 43.11, indicating that, on average, our model price predictions were off by $43.11. The model achieved a Root Mean Squared Error (RMSE) of $62.87, indicating that predicted nightly prices were, on average, within approximately $62.87 of actual prices. The model's R-squared value of 0.452 suggests that about 45.2% of the variance in Airbnb rental prices can be explained by the factors "neighborhood group," "neighborhood," "room type," and "availability." These values suggest a relatively strong model fit for a linear approach, considering the complexity and variability of Airbnb pricing.

Based on the performance of our model, we can draw several conclusions. First, while the model predictions are close to the actual prices for most listings, they're still off by an average of $43 per night. Second, RMSE is greater than MAE, suggesting some higher error predictions (likely due to expensive listings). Finally, although our model accounts for about 45% of the price variance, the remaining 55% of the price variance is likely derived from unobserved features (e.g., host reviews, seasonal trends, and amenities).

<u>Scatter Plot</u>

The Actual vs. Predicted Prices scatter plot visually illustrates that the model predicts low-to mid-range prices reasonably well, while higher-priced listings show larger deviations from the trend line.

Residual Plot

The residual plot further supports this pattern, as residuals widen at higher predicted prices, which indicates increasing underestimation for expensive listings.

These patterns align with the observed higher RMSE relative to MAE, suggesting that a small subset of high-priced outliers is contributing disproportionately to the model's overall error. Overall, the visual diagnostics corroborate the model metrics and highlight where prediction performance weakens.

6.2. Random Forest Regression

To further evaluate nonlinear modeling approaches, we next implemented a random forest regression model additionally with using the same set of variables: borough, neighborhood, room type, and availability as independent variables, and price as the dependent variable. Random forest is an ensemble learning technique that combines the prediction results of multiple decision trees and allows it to capture complex interactions that linear models may overlook and nonlinear relationships between variables.

Random Forest Regression Model Performance

```
MAE: 43.3830535484126
MSE: 4125.402182059614
RMSE: 64.22929380010038
R-Squared: 0.42790233742973693
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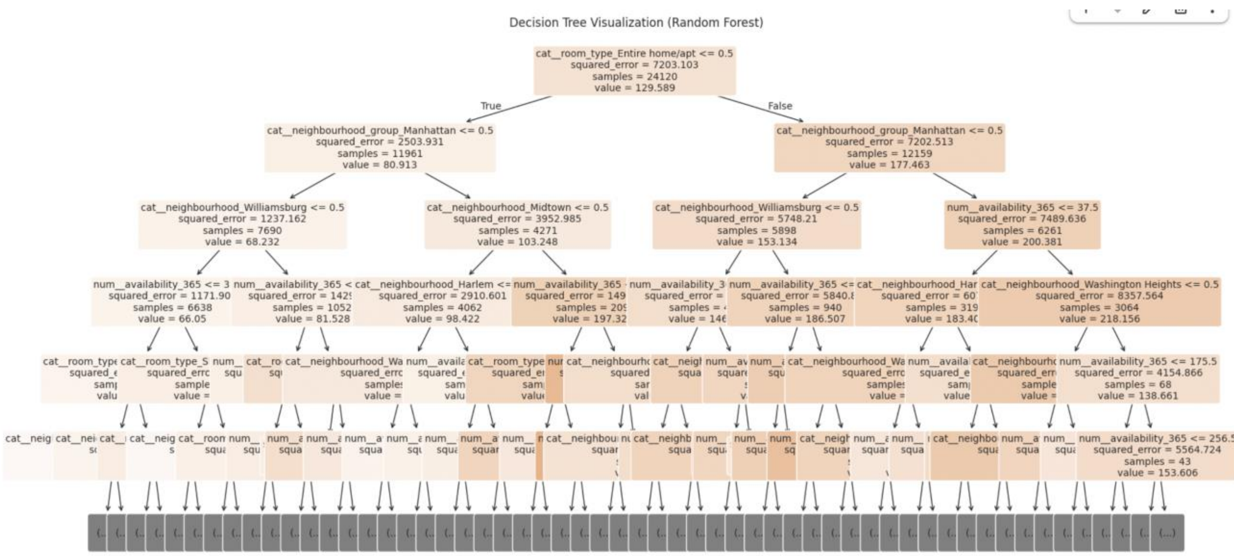
The random forest regression achieved a Mean Absolute Error (MAE) of 43.38, indicating that predicted nightly prices deviated from actual prices by approximately $43.38 on average. The model produced a Root Mean Squared Error (RMSE) of 64.23, suggesting the presence of larger prediction errors for certain listings, particularly higher-priced properties. The R-squared value of 0.428 indicates that approximately 42.8% of the variance in Airbnb rental prices can be explained by the model.

Overall, the random forest model performed slightly worse than the linear regression model across all evaluation indicators. Despite its ability to capture nonlinear relationships and feature interactions, Random Forest did not improve predictive accuracy in this setting. This suggests that, given the selected features, Airbnb pricing patterns in New York City may be largely linear or additive, and that the additional model complexity did not translate into better generalization performance.

Decision Tree: Random Forest



6.3. XGBoost Regression Analysis

To further improve predictive performance, we applied an XGBoost regression model, a gradient boosting framework that builds decision trees sequentially while correcting errors from previous iterations. This approach is particularly effective at capturing localized nonlinear patterns and subtle feature interactions, especially in structured tabular data.

Random Forest Regression Model Performance

```
MAE: 41.72980880737305
MSE: 3799.16064453125
RMSE: 61.63733158185265
R-Squared: 0.4731444716453552
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The XGBoost model achieved the best overall performance among the three models. It produced a Mean Absolute Error (MAE) of 41.73, indicating the lowest average prediction error across models. The Root Mean Squared Error (RMSE) was 61.64, suggesting improved robustness to larger errors compared to both linear regression and Random Forest. Additionally, the model achieved the highest R-squared value of 0.473, meaning that approximately 47.3% of the variance in Airbnb rental prices was explained by the model.

While the improvement over linear regression is modest, it indicates that nonlinear and localized effects, particularly at the borough level do contribute to Airbnb price variation. XGBoost's sequential learning approach allows it to capture these subtle patterns more effectively, making it the most accurate model for price prediction in this analysis.

7. Results and Conclusions

Based on the performance of the three models evaluated - linear regression, random forest regression, and XGBoost regression - we conclude that while simpler models are able to predict lower- to mid-priced Airbnb listings reasonably well, the overall pricing structure of Airbnb listings in New York City exhibits patterns that benefit from more flexible modeling approaches. In particular, high-priced listings display nonlinear and localized price dynamics that are not fully captured by a linear regression model.

The comparative results show that incorporating tree-based and boosting methods improves predictive performance, with XGBoost achieving the strongest overall results among the models considered. This suggests that Airbnb pricing in New York City is influenced not only by additive linear effects, but also by more complex relationships at finer geographic levels. For future analysis, to reduce RMSE and enhance model performance we can incorporate richer variables - such as amenities, host characteristics, seasonal factors, and review metrics - particularly for high-priced listings. For seasonal effects we could further look at weekend vs weekday pricing, and holiday pricing, which can help when explaining the short-term demand fluctuations, especially in Manhattan that is popular among tourists. In addition, to avoid high-priced outliers to dominate RMSE, using segmented modeling, such as modeling luxury listings separately – can improve performance. By using k-fold cross-validation, we could improve robustness and limit the sensitivity by training and evaluate the model for many various subsets of data.

8. Practical Implications

In conducting this analysis, we provide Airbnb platform, hosts, travelers, New York City policymakers, and investors, with a comprehensive breakdown of how Airbnb property characteristics influence pricing across the city.

This analysis helps policymakers and urban planners better understand how short-term rental activity affects the whole city. Our findings show that listings have a strong influence, especially in heavy tourist areas, such as Manhattan. This information can help city leaders make more informed decisions about policies in a city where the cost and availability of housing for NYC residents, especially those who have lived there for generations, has become a major issue of the 2020s.

For hosts, this analysis improves transparency regarding the factors that drive listing prices. Our findings indicate that location, including borough, neighborhood and room type, are the strongest factors for Airbnb pricing. It also suggests that hosts in high-demand neighborhoods, such as Manhattan, have less room to compete with prices alone and could instead focus on other features as well. In addition, the models show how high-priced listings can be underpredicted. This suggests that hosts in the segment also need to rely on other services and branding to show quality to justify the high prices.

This insight, in turn, empowers hosts to set competitive (and appropriate) prices to both attract customers and maximize profitability. For customers, the result means greater variety in their accommodation choices, in addition to greater transparency regarding the factors that affect Airbnb pricing.

If travelers want to find affordable options, this can be done by changing neighborhoods and room type, such as looking outside Manhattan that offer lower prices. If travelers also want to make more informed decisions, they should focus on listings with prices for low-to mid—range since our models show they are more predictable than the higher-priced listings.

Furthermore, the model performance can be used by Airbnb and similar platforms to improve pricing recommendation features for hosts. If using more demand indicators such as seasonality, this can support hosts in the app to know how much to price their listings, which in return will lead to more bookings for the Airbnb platform. Additionally, observing the sensitivity of RMSE to those high-priced outliers can support having a tiered pricing tool and keep the luxury listings separately to better support the hosts and guide them.

This analysis also offers valuable insights for stakeholders and market analysts seeking to navigate and manage the short-term rental market in the city more effectively. Our findings show that pricing variability increases substantially at higher price levels. This results in more risk and uncertainty for the luxury short-term rental market and suggests that investors should also be aware of the unobserved factors, such as host reputation when evaluating return. The XGBoost performance also suggests that analysis for investment should be done at neighborhood level instead of city-wide averages.

9. References

1. Dgomonov. “New York City Airbnb Open Data.” *Kaggle*, Kaggle, Aug. 2019, www.kaggle.com/datasets/dgomonov/new-york-city-airbnb-open-data.

2. Stringer, Scott M. “The Impact of Airbnb on NYC Rents.” *Office of the NYC Comptroller*, The City of New York, Apr. 2018, comptroller.nyc.gov/wp-content/uploads/documents/AirBnB_050318.pdf.